

Total no. of Pages:

Roll no.

6th Semester**MID-SEMESTER EXAMINATION, FEBRUARY-MARCH, 2026**

Course Title: Optimization Techniques in AI & ML

Course Code: EAEPE603

Time: 1.5 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	What is optimization? Why is it essential in engineering and decision-making problems?	2	CO1
1b	On what basis are optimization problems classified? Discuss the classification with examples.	2	CO1
2a	What does the No Free Lunch Theorem imply about optimization algorithms?	2	CO2
2b	What is a Metaheuristic Algorithm? Compare Heuristic and Metaheuristic Algorithms.	2	CO2
3a	Explain simulated annealing algorithm while mentioning its way for exploration and exploitation.	2	CO2
3b	How do GA, PSO, and SA differ in their approach to Exploration and Exploitation?	2	CO2
4a	What are the hyperparameters involved in Particle Swarm Optimization (PSO)? Outline the steps of the algorithm.	2	CO2
4b	Define the Crossover and Mutation operators in Genetic Algorithms with examples, and explain the role of crossover and mutation rates in generating the next generation. Or Define Learning Rate and Saddle Point in the context of Gradient Descent, and what happens if the learning rate is too small or too large?	2	CO1
5a	What are common variants of gradient descent, and how do they differ from each other?	2	CO1
5b	A dataset contains 100 training samples. The model is trained using a batch size of 20 for 5 epochs. (i) Calculate the number of iterations per epoch. (ii) Determine the total number of iterations during training. Consider the cost function $J(w) = (w-3)^2$ Let the initial weight be $w_0=0$ and the learning rate be $\eta=0.2$. Find the updated weight using one iteration of Gradient Descent. Or Find the minimum value of the function using Simulated Annealing Minimize $f(x) = x_1^2 + x_2^2 - 5x_1x_2 - 20x_1 - 15x_2 + 400$ For $-3 < x_1 < 10$ and $-1 < x_2 < 10$; Assume temperature reduction factor $c=0.7$, Maximum permissible number of iterations as, $n=2$.	2	CO2